4.8

	A	B	$\bar{A}B$	$A\bar{B}$	$A \oplus B$	$\overline{A \oplus B}$
#0	0	0	0	0	0	1
#1	0	1	1	0	1	0
#2	1	0	0	1	1	0
#3	1	1	0	0	0	1

$y = A \oplus B$

$y : \mathcal{E}_2^2 \to \mathcal{E}_2$

$y = m(1,2)$

$A \oplus B = \bar{A}B + A\bar{B}$

$\overline{A \oplus B} = \bar{A}\bar{B} + AB$

XOR $A \oplus B = \bar{A}B + A\bar{B}$

XOR NEGAT $\overline{A \oplus B} = \bar{A}\bar{B} + AB$

$\bar{y} : \mathcal{E}_2^2 \to \mathcal{E}_2$

$\bar{y} = m(0,3)$

XOR =1

ŞI &

SAU ≥1

	A	B	C	D	P	i
#0	0	0	0	0	1	0
1	0	0	0	1	0	1
2	0	0	1	0	0	1
3	0	0	1	1	1	0
4	0	1	0	0	0	1
5	0	1	0	1	1	0
6	0	1	1	0	1	0
7	0	1	1	1	0	1
8	1	0	0	0	0	1
9	1	0	0	1	1	0
10	1	0	1	0	1	0
11	1	0	1	1	0	1
12	1	1	0	0	1	0
13	1	1	0	1	0	1
14	1	1	1	0	0	1
15	1	1	1	1	1	0

$P = \bar{C}\bar{D}(\bar{A}\bar{B} + AB) + \bar{C}D(\bar{A}B + A\bar{B}) + C\bar{D}(\bar{A}B + A\bar{B}) + CD(\bar{A}\bar{B} + AB)$

q :

AB \ CD	00	01	11	10
$\bar{C}\bar{D}$	1	0	1	0
$\bar{C}D$	0	1	0	1
CD	0	0	0	0
$C\bar{D}$	0	0	0	0

$q = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} + AB\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D}$

$= \bar{C}(\bar{A}\bar{B}\bar{D} + \bar{A}BD + AB\bar{D} + A\bar{B}D)$

$= \bar{A}\bar{C}(\bar{B}\bar{D} + BD) + A\bar{C}(B\bar{D} + \bar{B}D)$

$= \bar{A}\bar{C}(\overline{B \oplus D}) + A\bar{C}(B \oplus D)$

Cati de 1 am?

$P = m(0,3,5,6,9,10,12,15)$

$F_e C_p = M(1,2,4,7,8,11,13,14)$

AB \ CD	00	01	11	10
00	1	0	1	0
01	0	1	0	1
11	1	0	1	0
10	0	1	0	1

$= (\overline{A+B})(C+D) + (A+B)(C+D)$

$= (\overline{A+B})(\overline{C+D}) + (A \oplus B)(C \oplus D)$

$= \overline{(A+B) \oplus (C \oplus D)}$

$) = m(1, 2, 4, 7, 8, 11, 13, 14)$

$\bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}BC\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}CD + AB\bar{C}D + ABC\bar{D}$

4

$FCCi = M(0, 3, 5, 6, 9, 10, 12, 15)$

$= (A+B+C+D)(A+B+\bar{C}+\bar{D})(A+\bar{B}+C+\bar{D})$

$(A+\bar{B}+\bar{C}+D)(\bar{A}+B+C+\bar{D})(\bar{A}+B+\bar{C}+D)$

$(\bar{A}+\bar{B}+C+D)(\bar{A}+\bar{B}+\bar{C}+\bar{D})$

CD \ AB	00	01	11	10
00	0	1	0	1
01	1	0	1	0
11	0	1		1
10	1	0	1	0

$\bar{C}\bar{D}(\bar{A}B + A\bar{B}) + C\bar{D}(\bar{A}B + AB) +$

$+ \bar{C}D(\bar{A}B + AB) + CD(\bar{A}B + A\bar{B})$

$= \bar{C}\bar{D}(\bar{A}B + A\bar{B}) + C\bar{D}(\bar{A}B + AB) +$

$+ (\bar{A}B + AB)(\bar{C}D + C\bar{D})$

$= (A \oplus B)(\overline{C \oplus D}) +$

$+ \overline{(A \oplus B)}(C \oplus D)$

$= (A \oplus B) \oplus (C \oplus D)$

fractional digram

$0,03125_{(10)} \rightarrow 0,00001_{(2)}$

$0,002_{(4)}$

$0,03125 * 2 = 0,06250 \quad 0$

$0,06250 * 2 = 0,1250 \quad 0$

$0,1250 * 2 = 0,250 \quad 0$

$0,025 * 2 = 0,5 \quad 0$

$0,5 * 2 = 1,0 \quad 1$

$n = 2^{-5} = 2^{-5}(1+0)$

$\hat{e_i} = -5$

$offset = 2^{k-1} - 1$

$k = 11$

$\Rightarrow exp = offset + e_i =$

$= 1023 + (-5)$

$= 1018 = 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^1$

9 8 7 6 5 4 3 2 1 0

0 , 0 1 1 1 1 1 1 0 1 0

26.11.2025

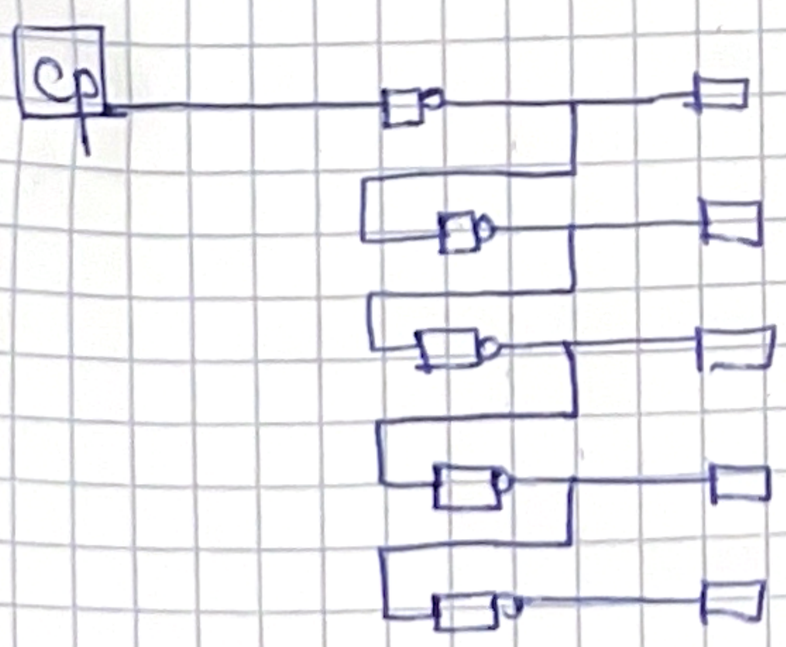
Gating

EN	D	y
0	0	0
0	1	0
1	0	0
1	1	1

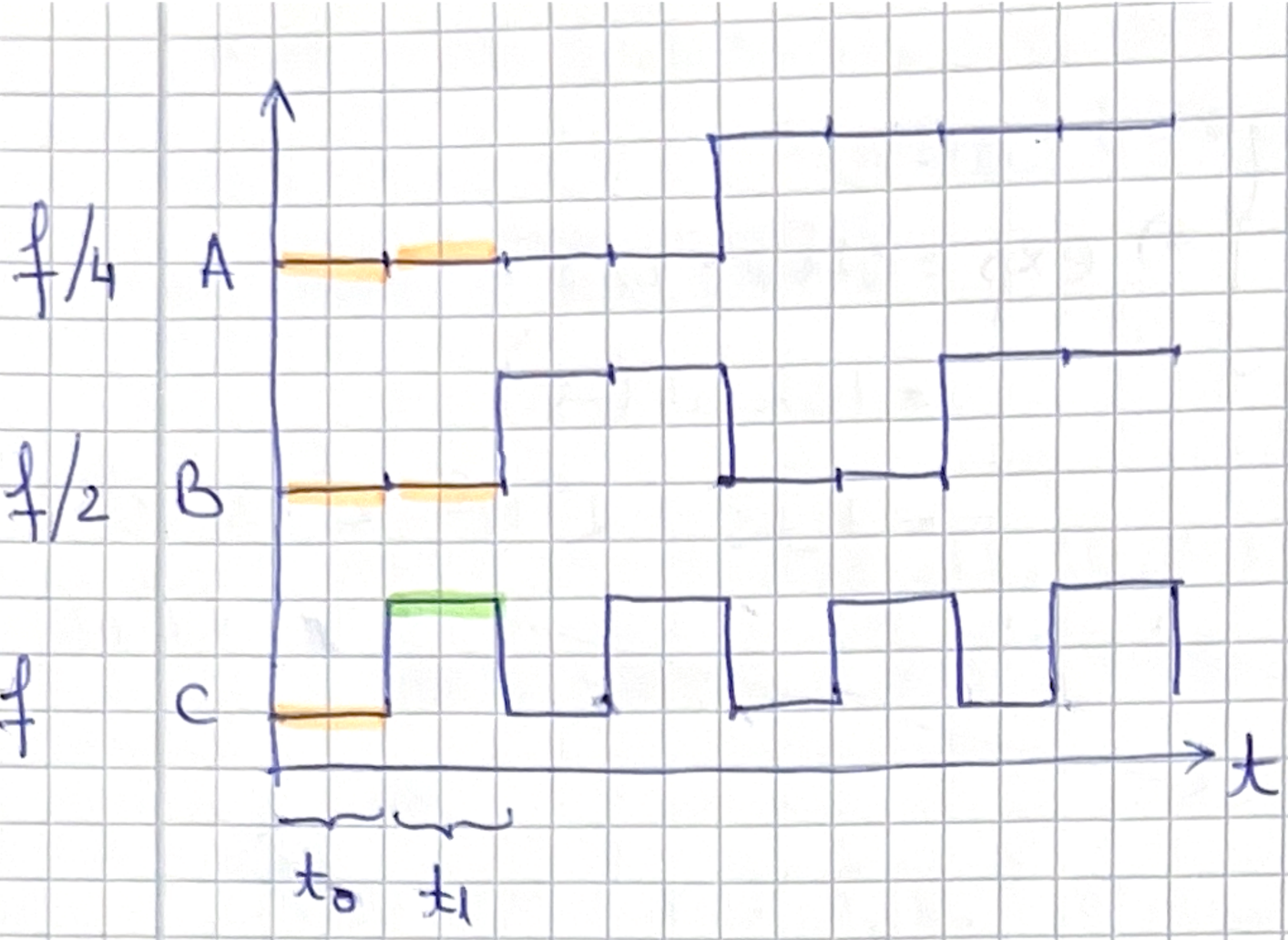
EN	D	y
0	x	0
1	0	0
1	↑	1

x - nu contează, adică nu schimbă semnătura $f_m(\neq CD)$ indiferent ce valoare are

Circuitul de întârziere



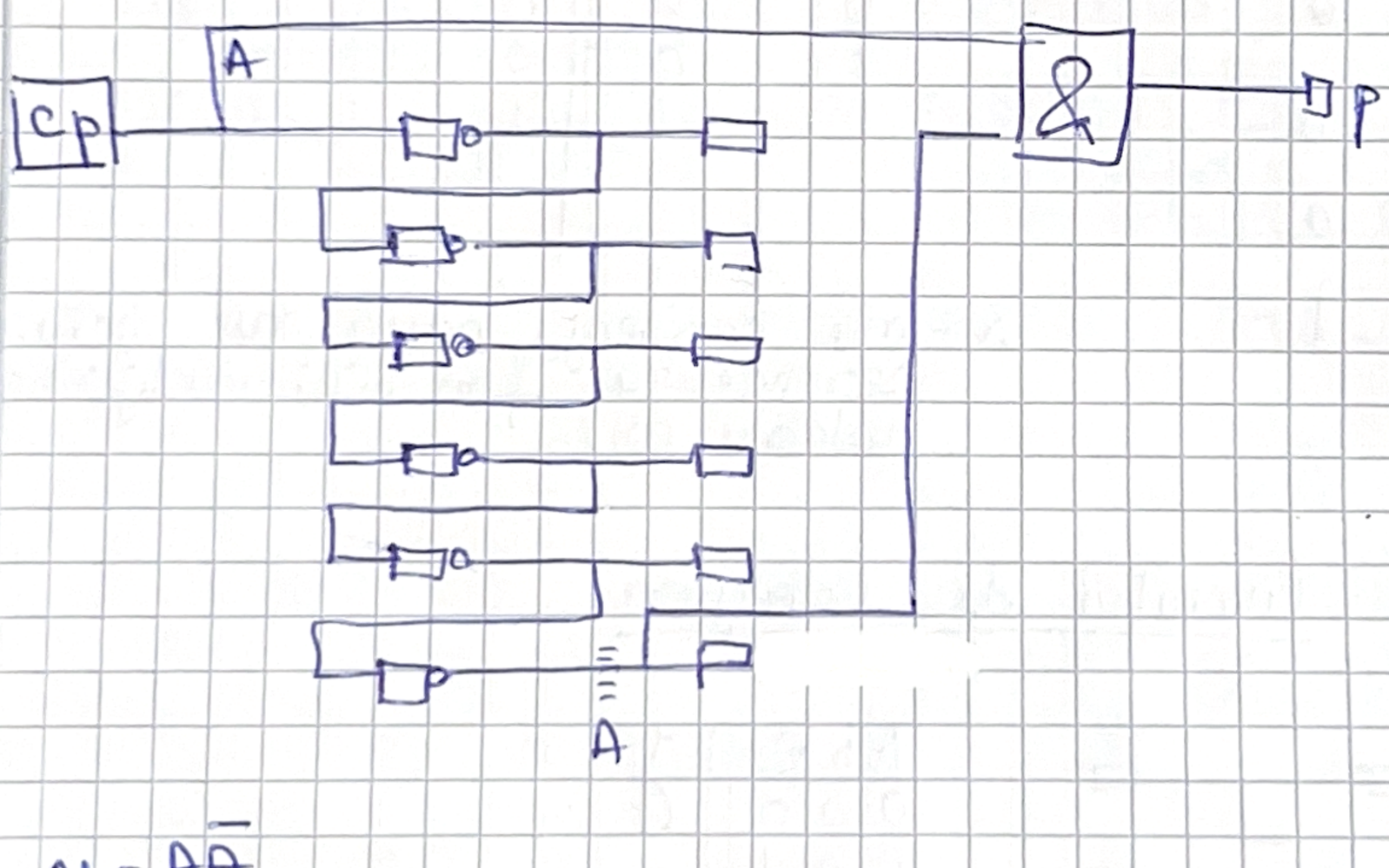
A	B	C	y
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	



$t_0 = t_1$ = clock

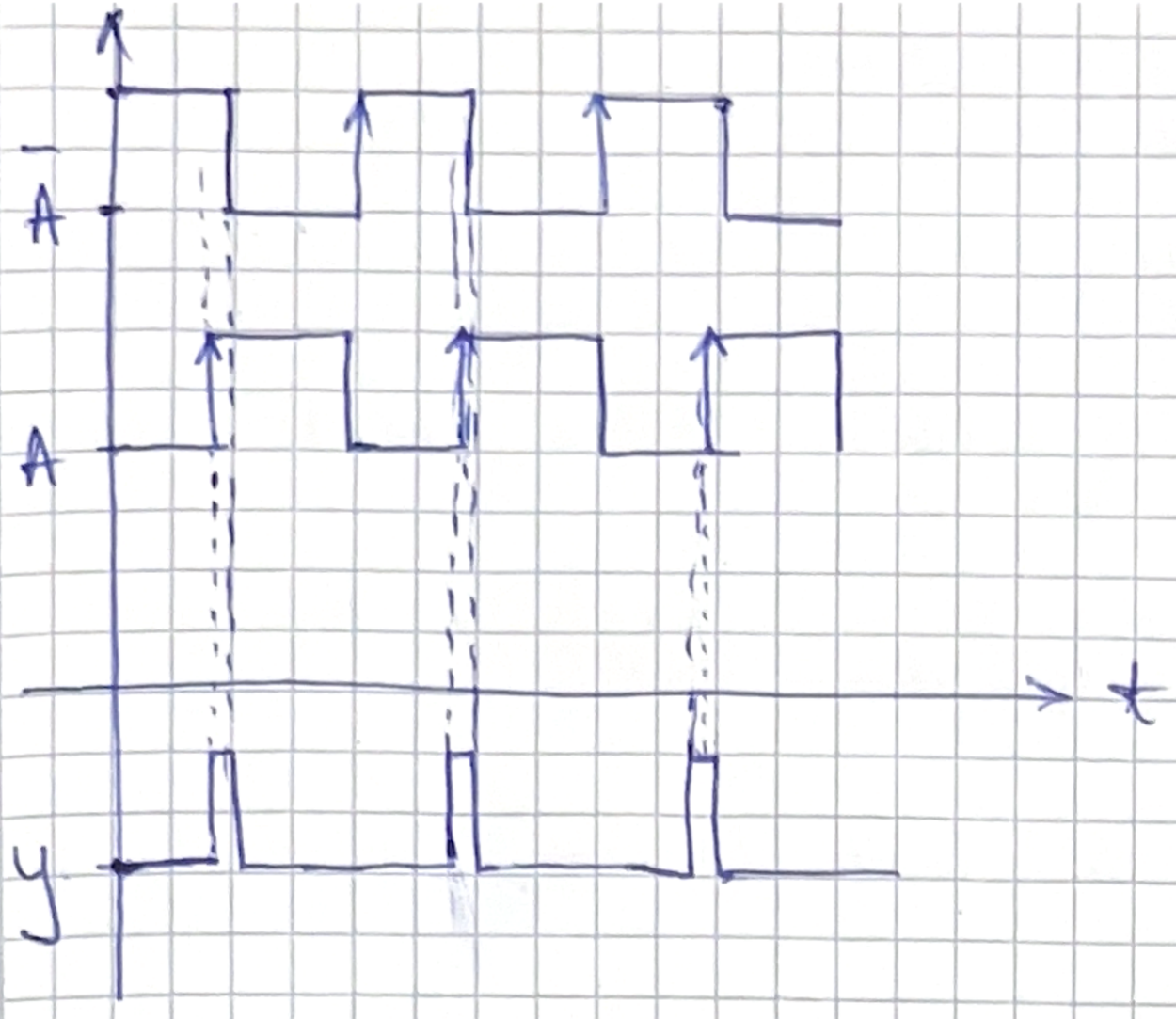
$t_0 <> t_1$ = puls

Detectarul de front ascendent - $A\bar{A}$

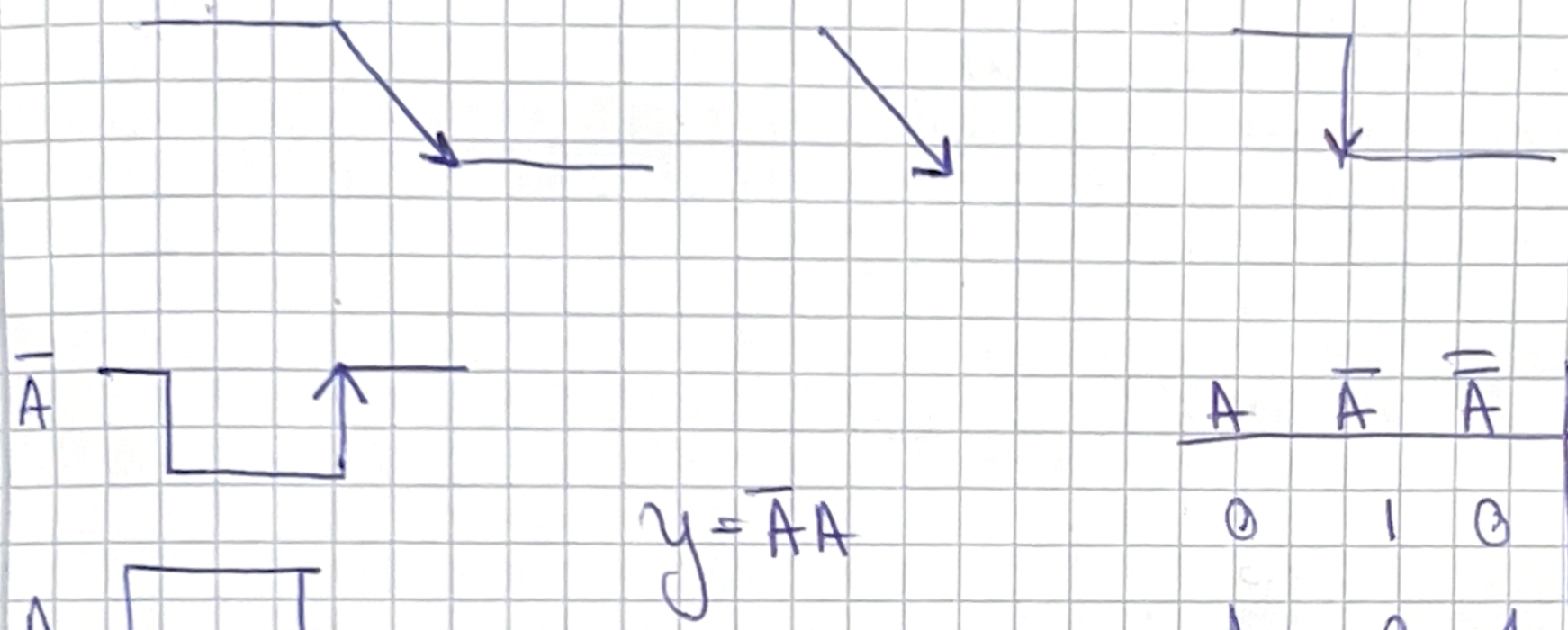


$y = A\bar{A}$

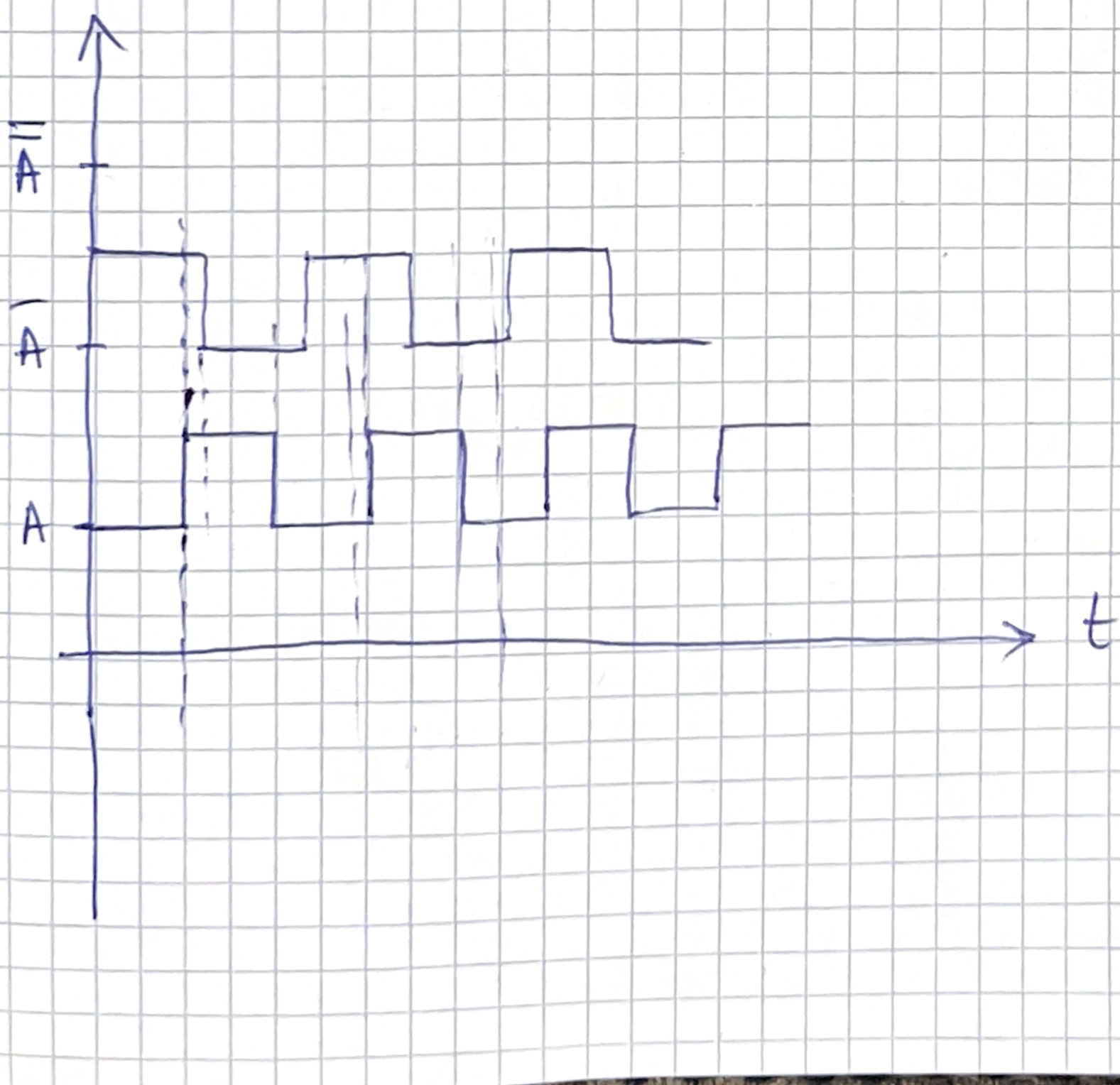
A	$\bar{A}$	y
0	1	0
1	0	0



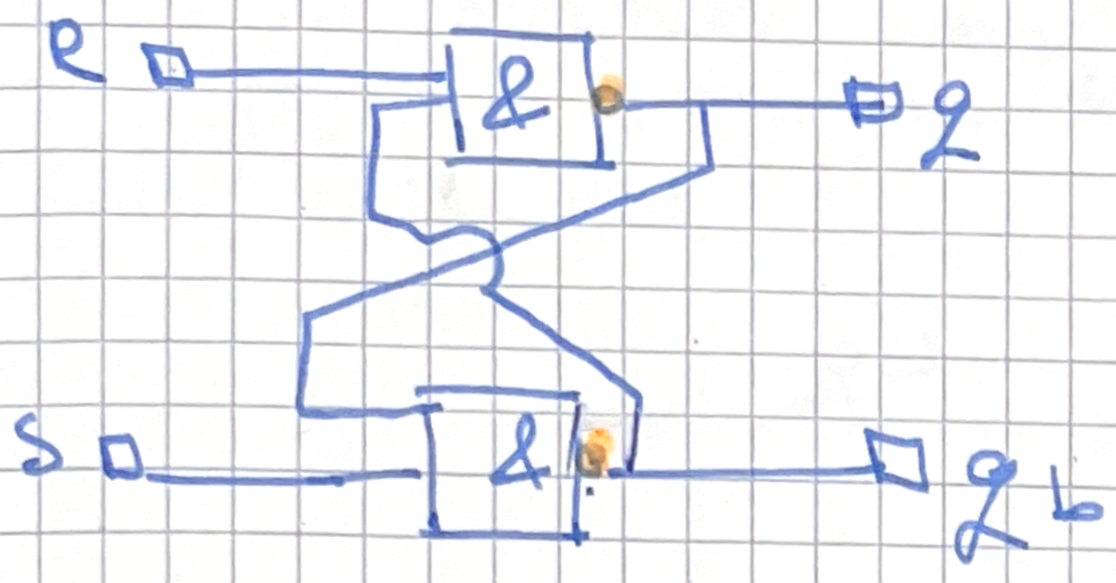
Detectorul de front descendent



A	$\bar{A}$	$\bar{\bar{A}}$	$\bar{A}\bar{\bar{A}}$
0	1	0	0
1	0	1	0



S-R NAND / NOR

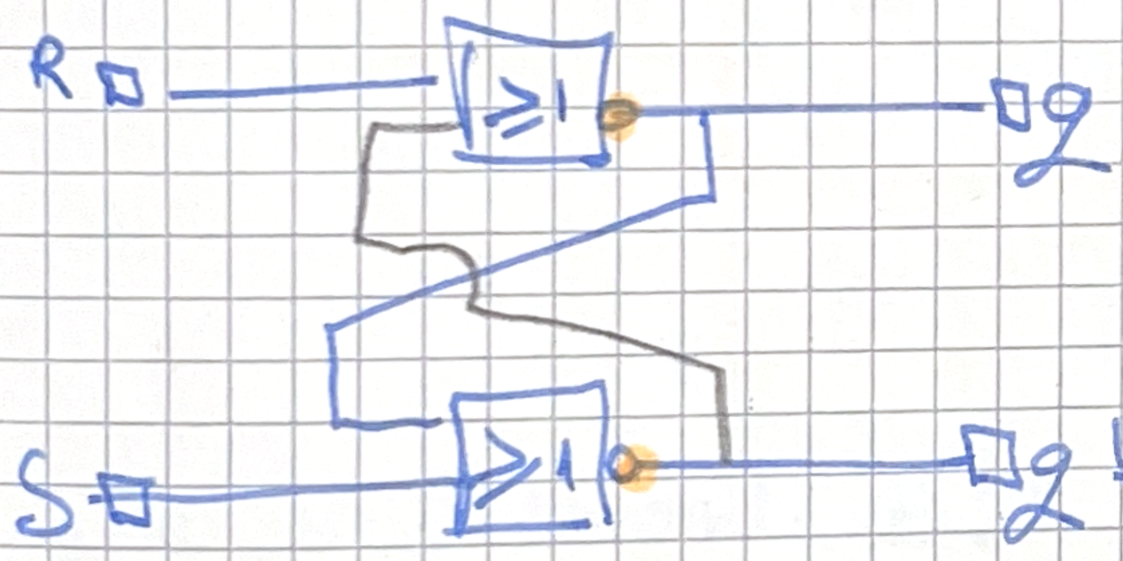


S-R NAND

R	S	Q	Q b	
0	0	1	1	
0	1	1	0	Set
1	0	0	1	RES
1	1	Q	Q b	Mem

$$Q = \overline{R \cdot Q_b} = \bar{R} + \overline{Q_b}$$

$$Q_b = \overline{S \cdot Q} = \bar{S} + \overline{Q}$$

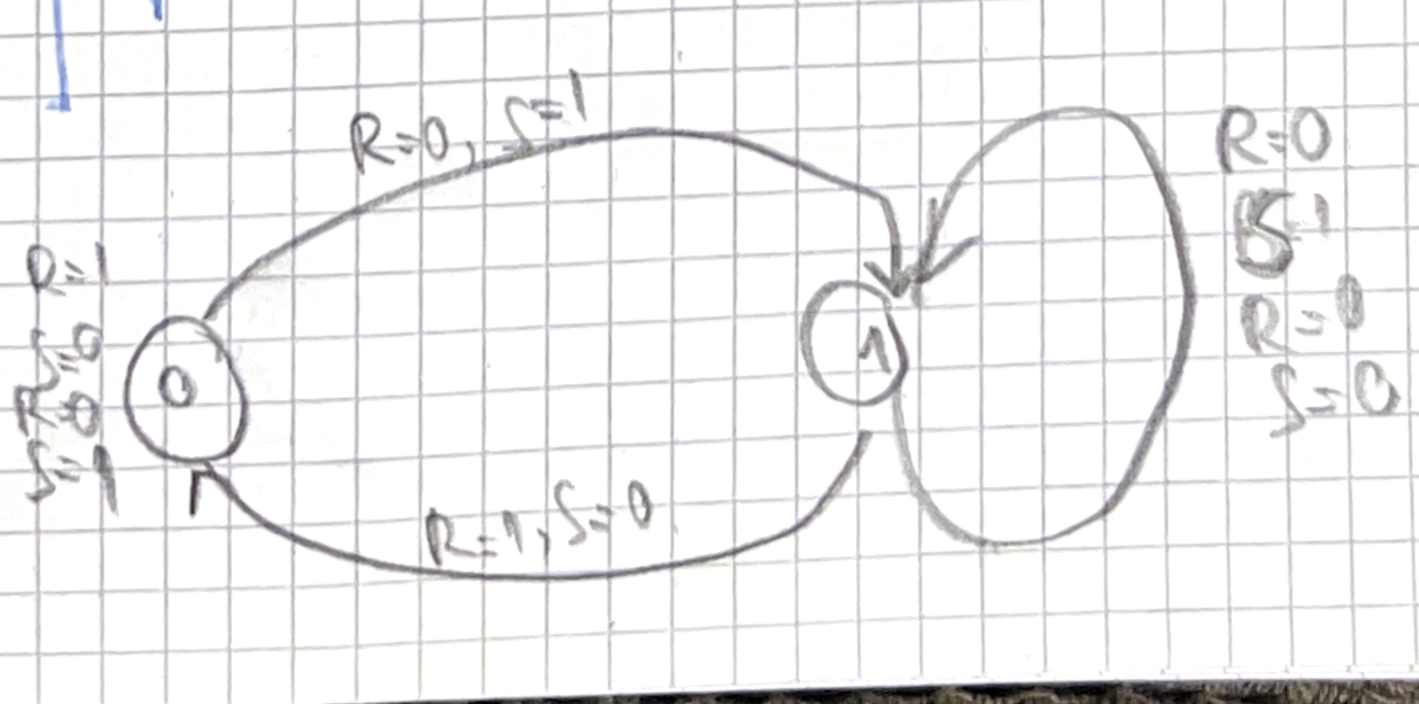


S-R NOR

R	S	Q	Qb	
0	0	Q	Qb	mem
0	1	1	0	set
1	0	0	1	Res
1	1	–	–	

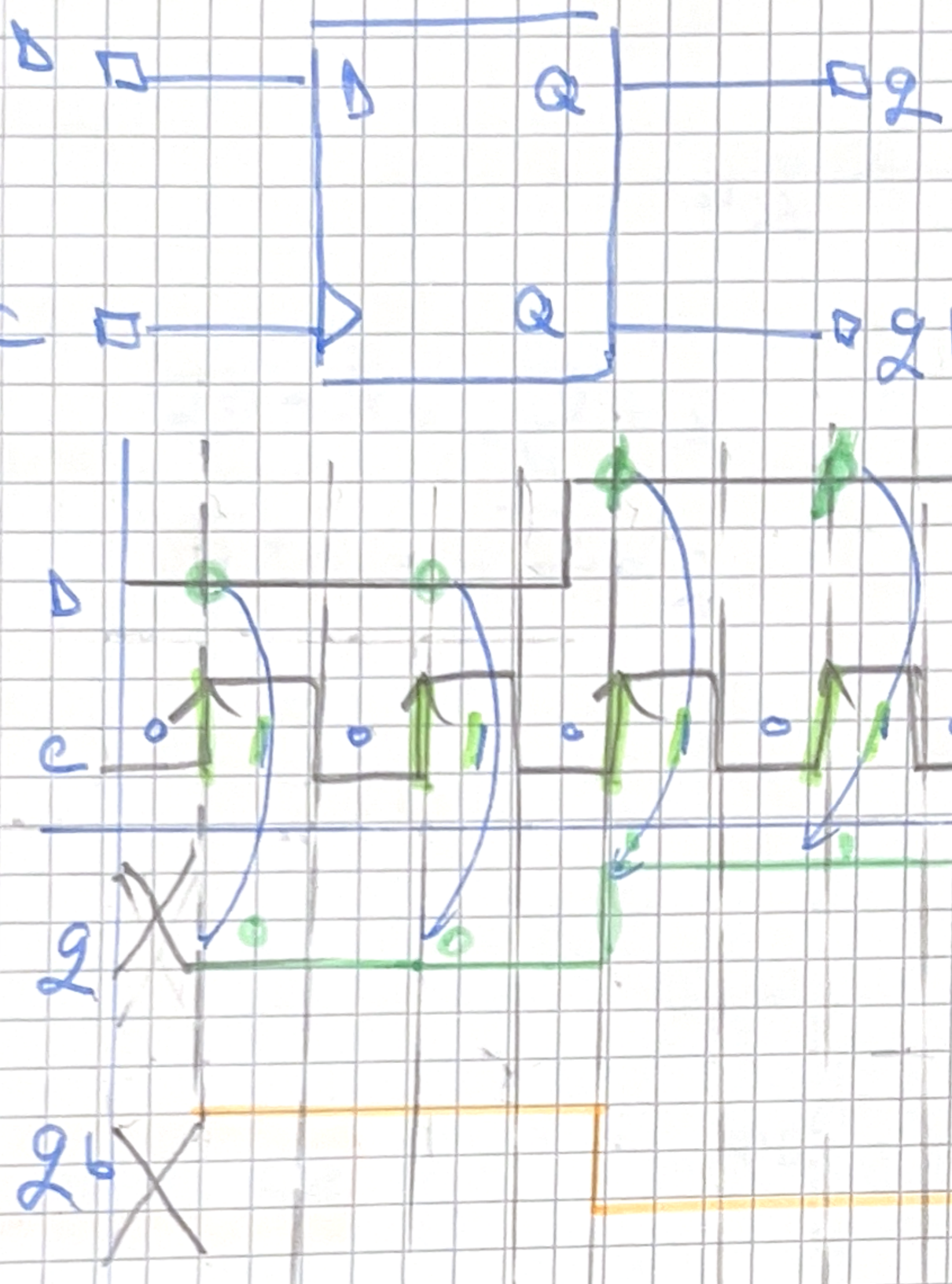
$$Q = \overline{R + Q_b} = \bar{R}\,\overline{Q_b}$$

$$Q_b = \overline{S + Q} = \bar{S}\,\overline{Q_b}$$



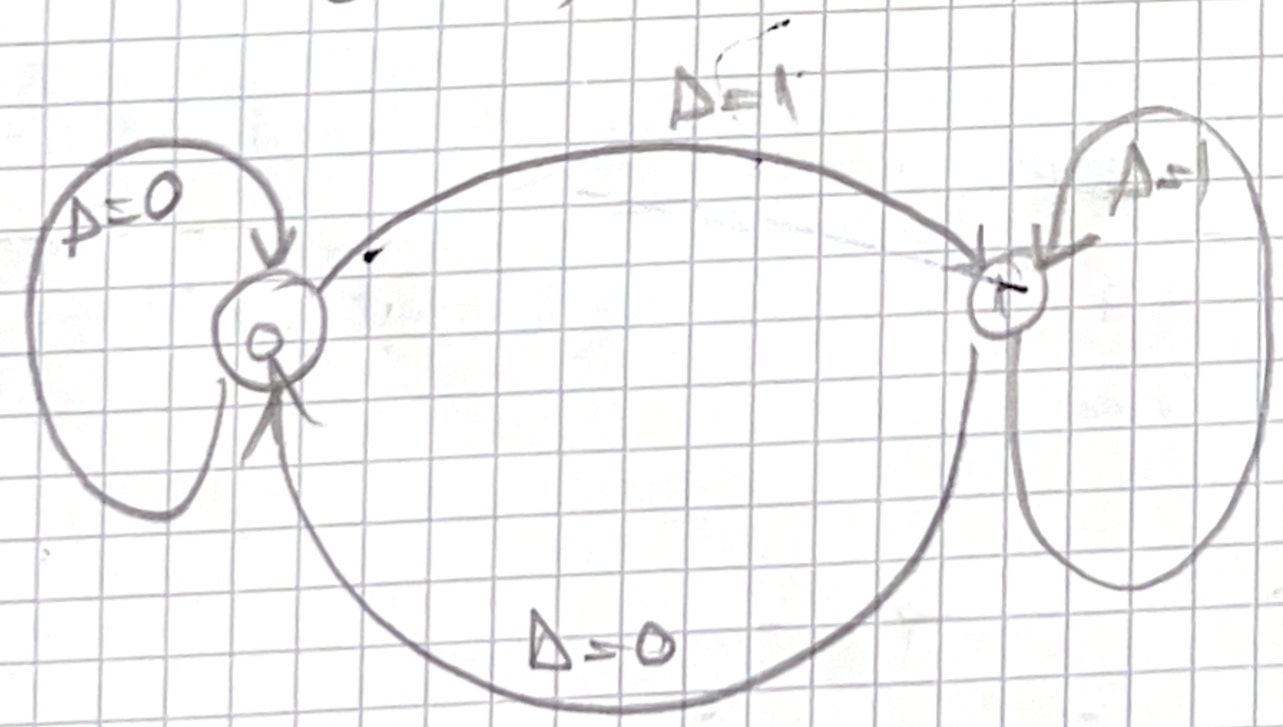
(Bistabil de tip D) BB D - Bistabil D - Copiază

Copiază D la Q pe frontul ascendent al semnalului.



MAȘINĂ de stări finite (FSM)

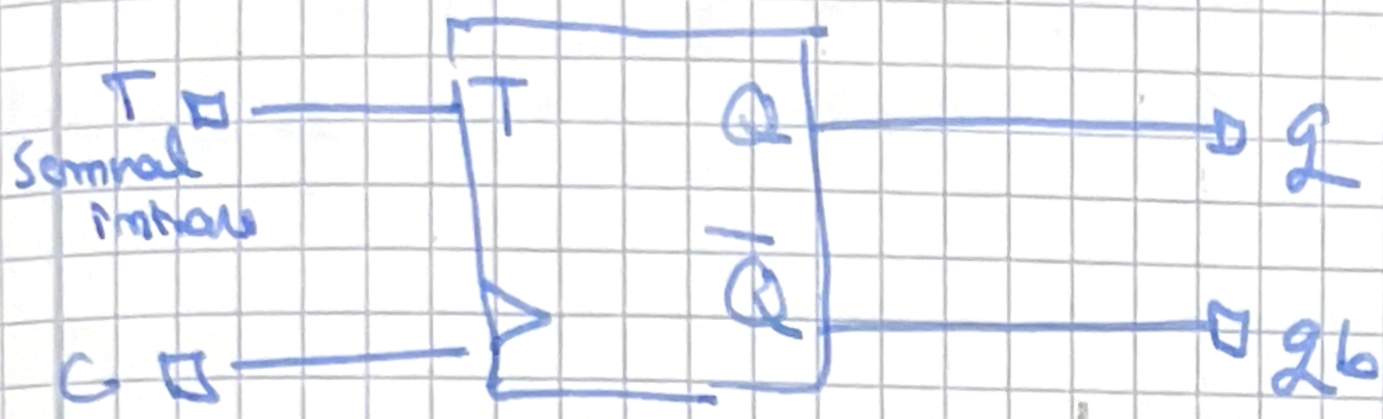
DCG : Directed Cyclic Graph

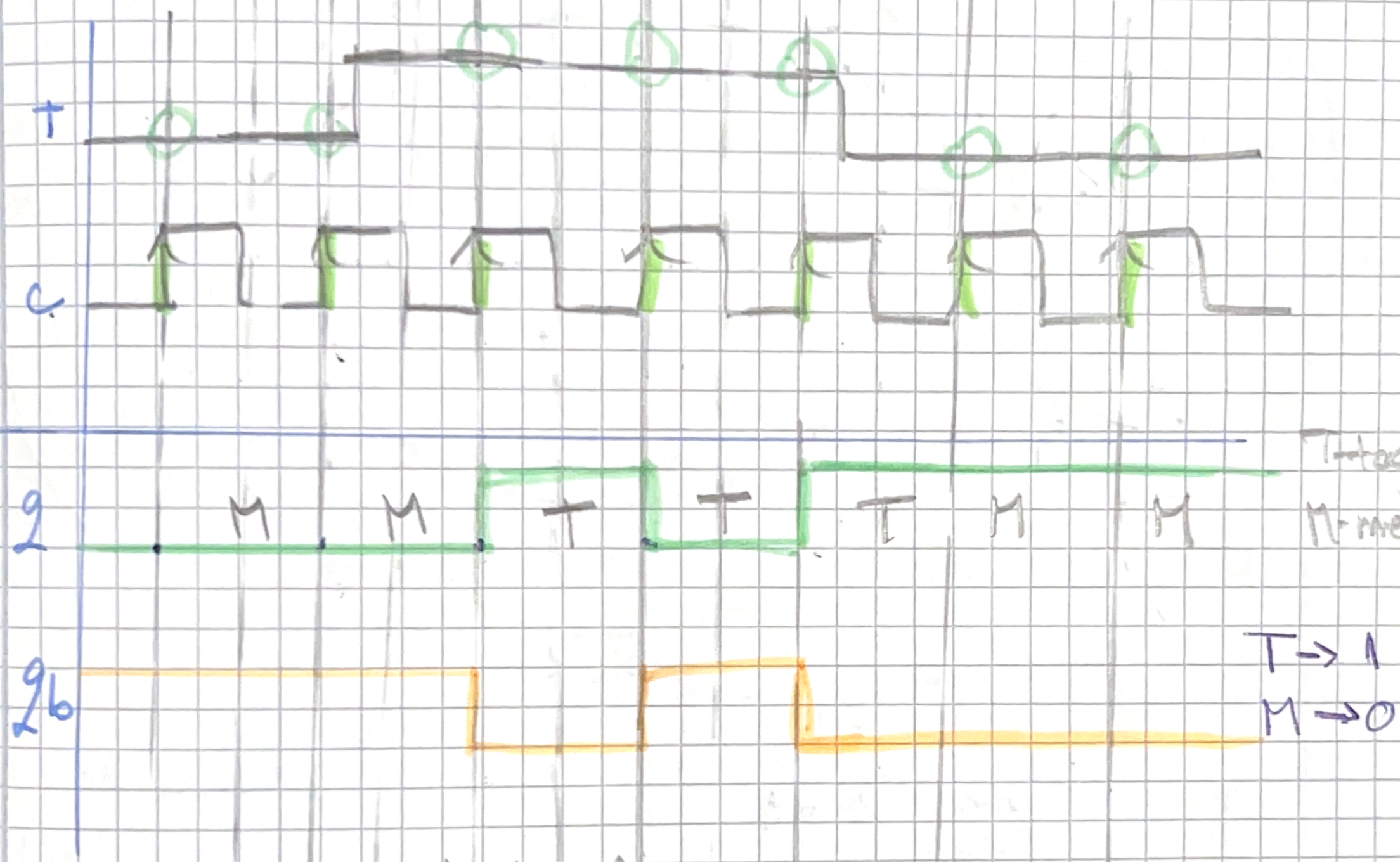


(bistabil de tip T) BB T

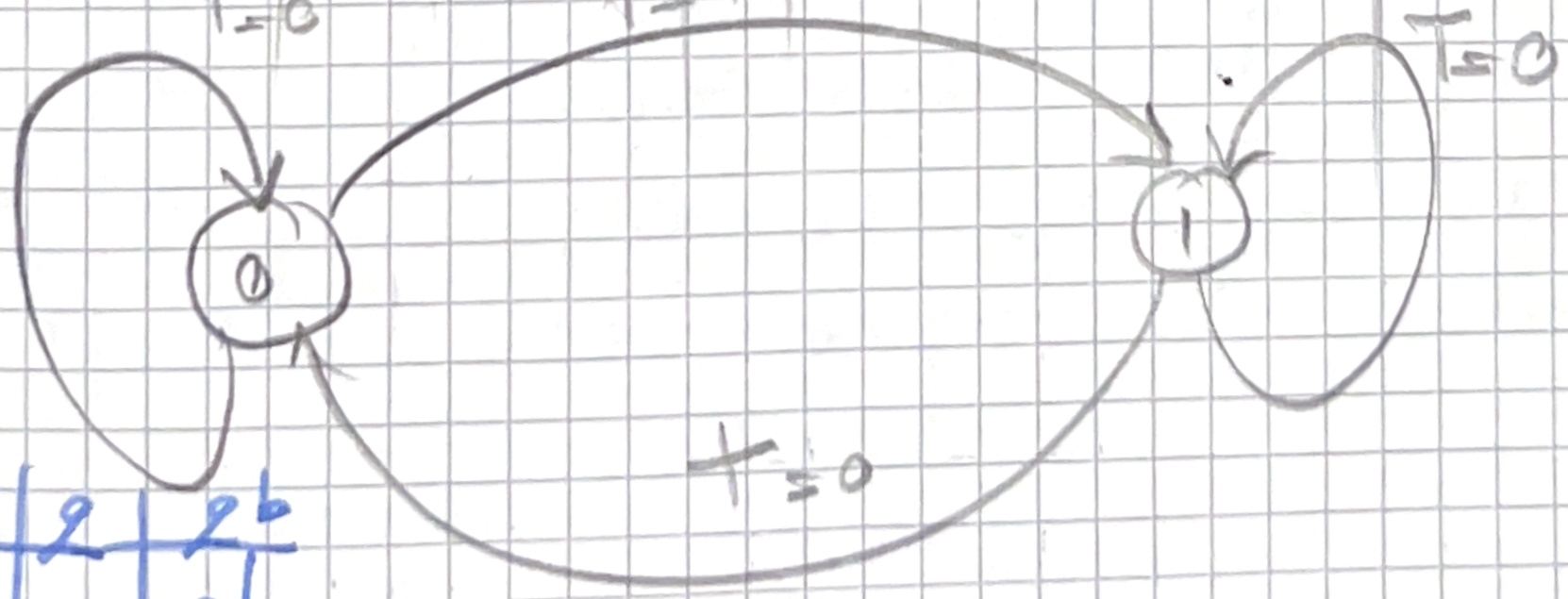
T - schimbă

Face toggle pe frontul ascendent al terminalului de ceas atunci când T=1.





Mașina pe stări finite



t	c	q	qb
0	x	q	qb
1	↗	qb	q

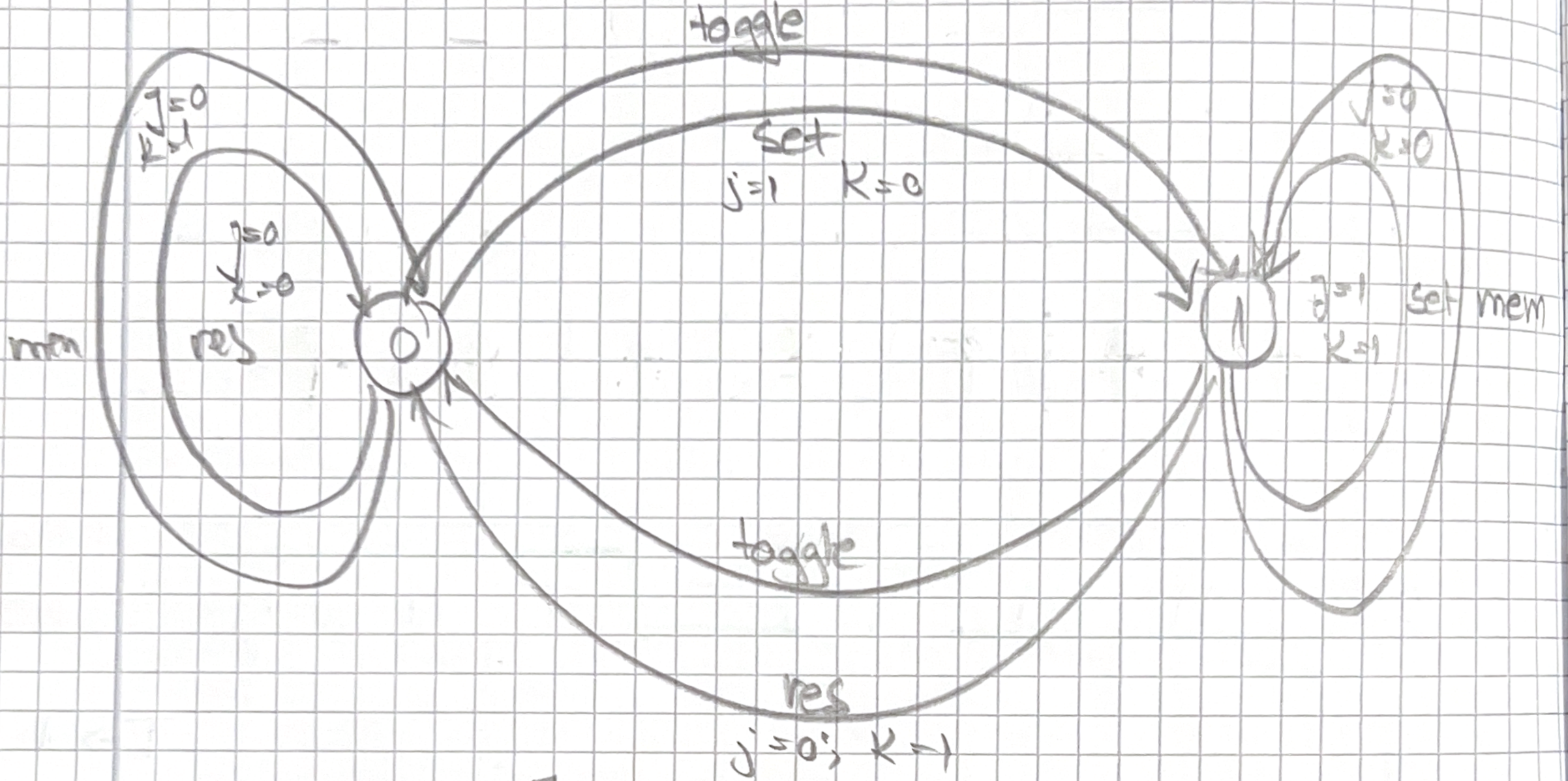
→ Mem

→ Toggle : schimbă între ele q cu qb

BB JK

Bistabil

Pe frontul ↑ al semnalului de ceas comparăm j la ieșirea Q dacă j⊕K=1, dacă j=K=0 face memorie
dacă j=K=1 face toggle



Toggle : $\begin{cases} J=1 \\ K=1 \end{cases}$

Tabela de excitație

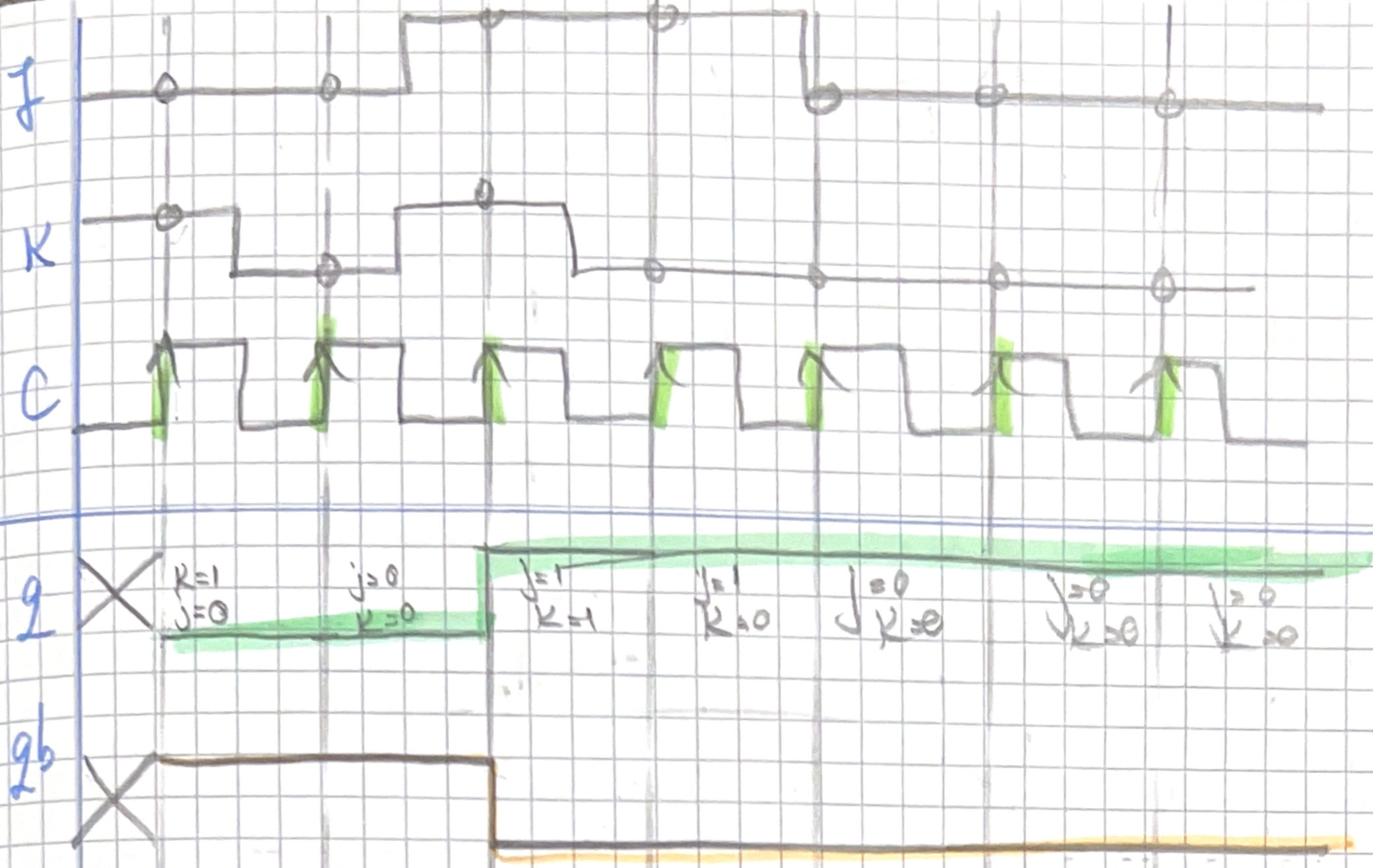
Q	Q_n	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

0: $J=0$; $K=x \longrightarrow 0$

0: $J=1$; $K=x \longrightarrow 1$

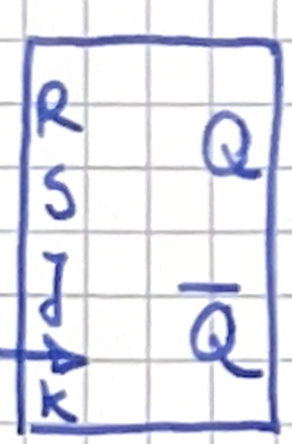
1: $J=0$; $J=x \longrightarrow 1$

1: $J=1$; $J=x \longrightarrow 0$

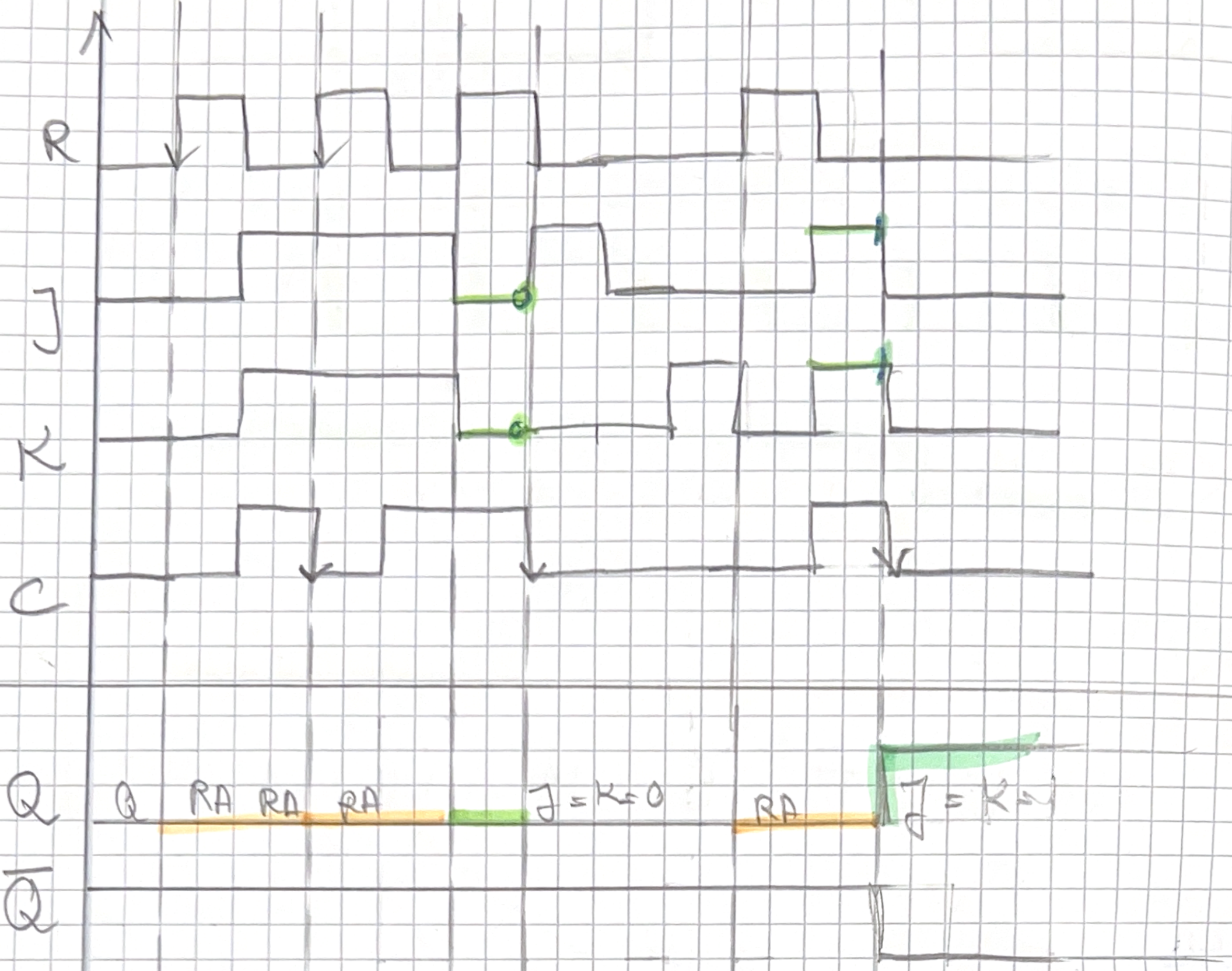


LU

bb JK -RJKC -08HE 19102870 , Qi = 0



#	R	J	K	C	Q	Q̄	
0	0	0	0	0	0	1	Q
8	1	0	0	0	0	1	RA
7	1	0	1	1	0	1	M
E	1	1	1	0	0	1	RA
7	1	0	1	1			M
9	1	0	0	1	0	1	RA
4	0	1	0	0			MS (memorare simcon)
0	0	0	0	0			M
2	0	0	1	0			M
8	1	0	0	0	0	1	RA
7	0	1	1	1			M
0	0	0	0	0			Toggle

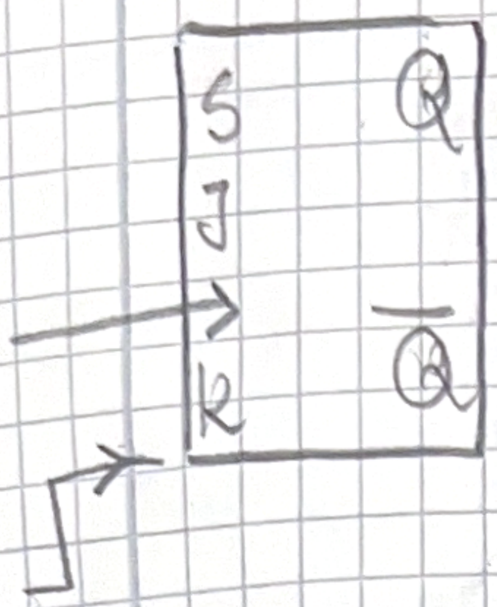


0 8 4 E 7 9 4 0 2 8 7 0

1. Semirate asincrom
2. MS / Toggle

BBJK + SJKC - 3003402C1928

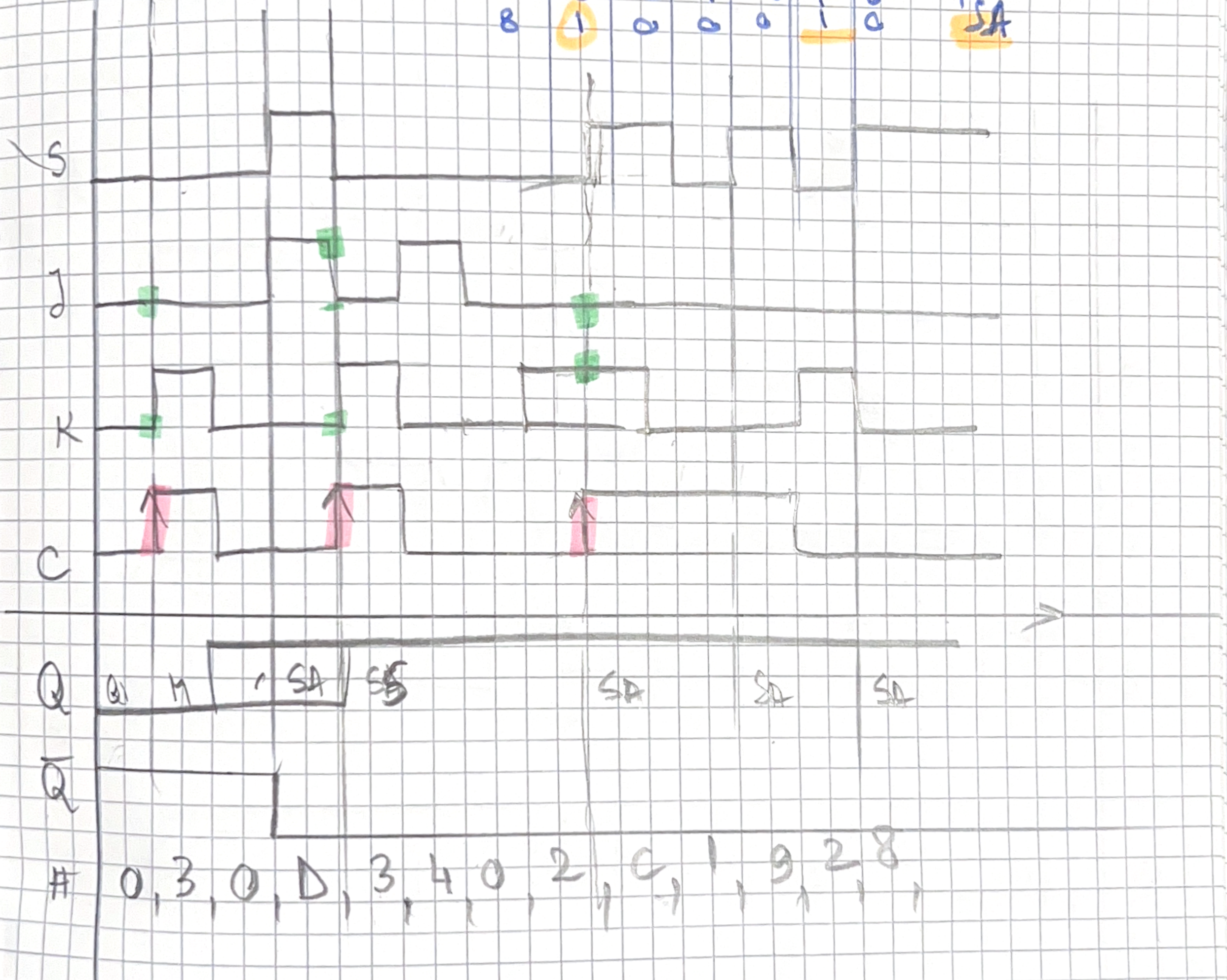
$Q_i = 0$



#	S	J	K	C	Q	$\overline{Q}$
3	0	0	1	1	0	1
0	0	0	0	0	0	1
D	1	1	0	0	1	0
3	0	0	1	1	0	1
4	0	1	0	0	1	0
0	0	0	0	0	1	0
2	0	0	1	0	1	0
C	1	0	1	1	1	0
1	0	0	0	1	1	0
9	1	0	0	1	1	0
2	0	0	1	0	1	0
8	1	0	0	0	1	0

Q
H
SA
SS
H
H
K
SA
H
JA
H
JA

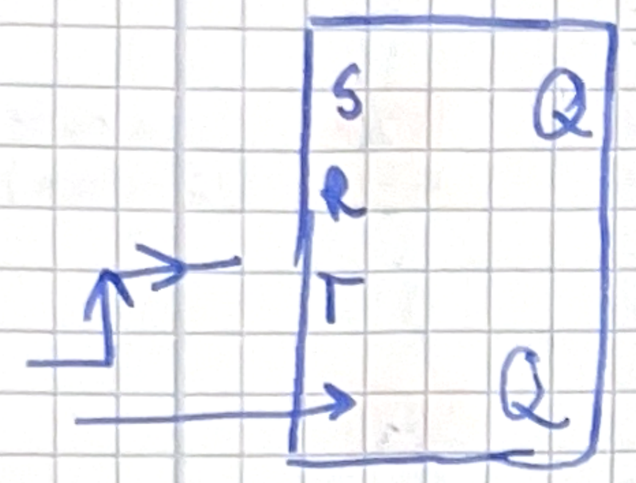
(asincron)
(sincron)



SA - ignorăm C

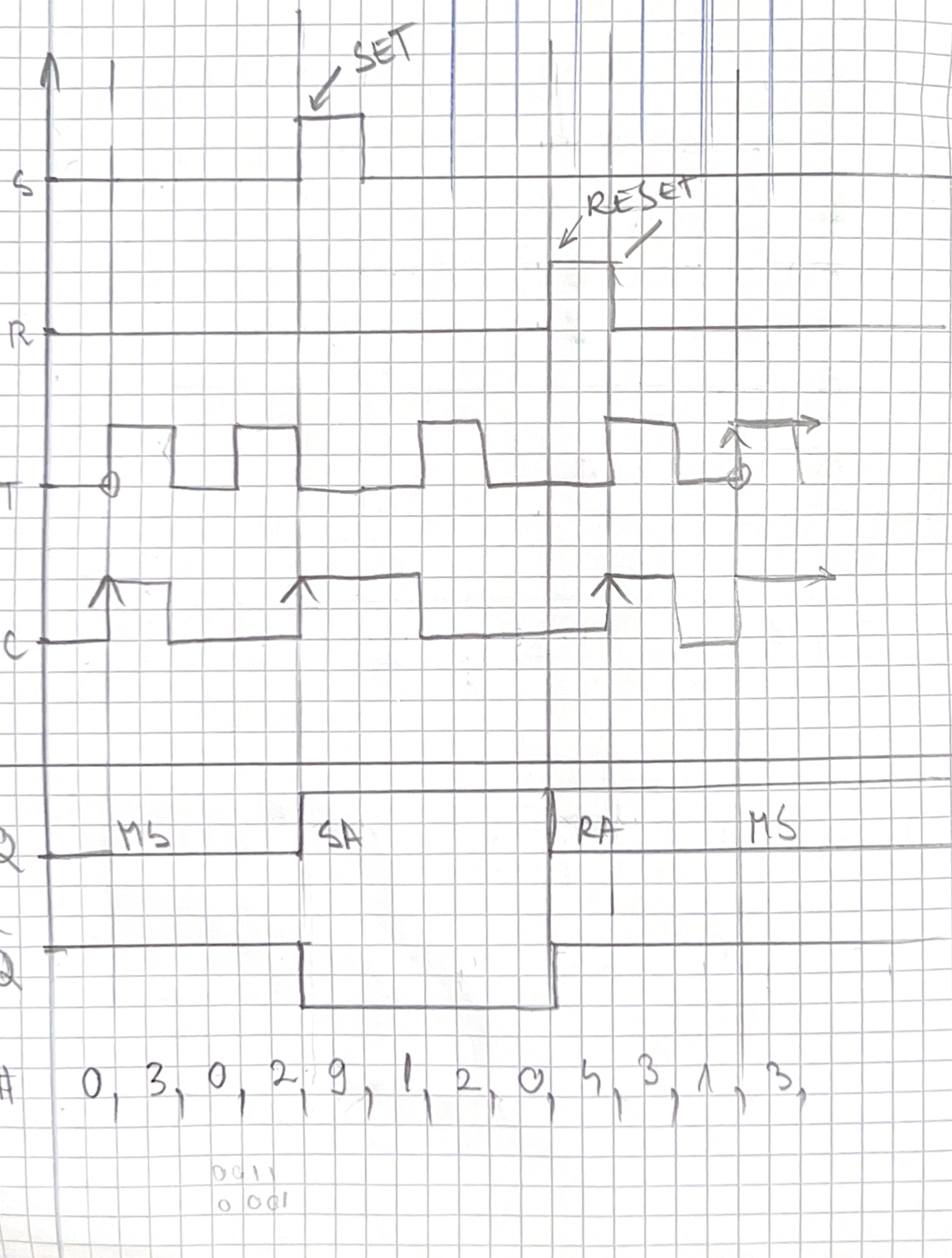
BB T + SETC = 03029120431 3

$\overline{x} = X$



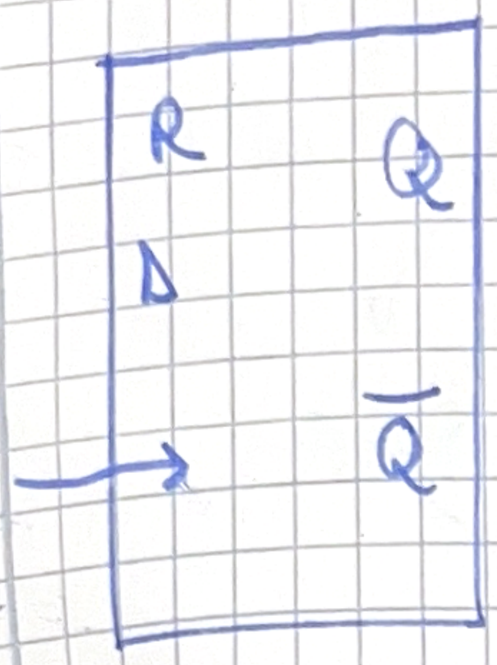
#	S	R	T	C	Q	$\overline{Q}$	
0	0	0	0	0	0	1	Q_i
3	0	0	1	1			MS
0	0	0	0	0			MS
2	0	0	1	1	1	0	SA
9	1	0	0	1			MS
1	0	0	1	0			MS
2	0	0	0	0			MS
0	0	1	0	0	0	1	RA
4	0	0	1	1			MS
3	0	0	0	0			MS
1	0	0	1	1			MS
3	0	0	0	0			MS

$Q_i = 0$

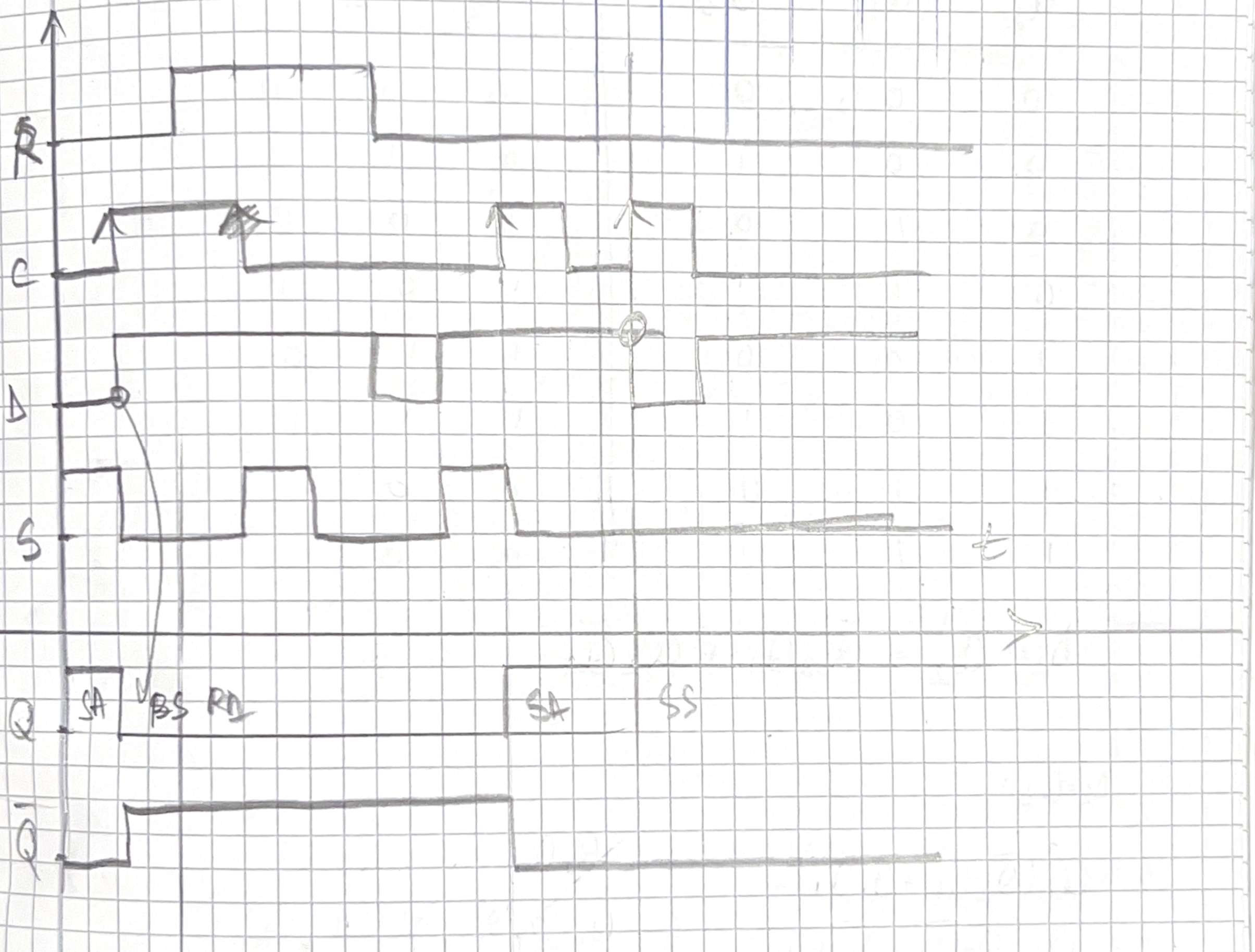


0, 3, 0, 2, 9, 1, 2, 0, 4, 3, 1, 3,

BB D + RCJS 16EBA036242 Qi =?, x



#	R	C	D	S	Q	Q̄	
1	0	0	0	1	x	x	SA
6	0	1	1	0	0	1	RS
E	1	1	1	0	0	1	RA
B	1	0	1	1	0	1	RA
A	1	0	1	0	0	1	RA
0	0	0	0	0	0	1	M
3	0	0	1	1	1	0	SA
6	0	1	1	0	1	0	SS
2	0	0	1	0	1	0	M
4	0	1	0	0	1	0	SS
2	0	0	1	0	1	0	M



1, 6, E, B, A, 0, 3, 6, 2, H, 2,